

Introduction

We wanted to solve a problem that a person/organization already had. We wanted to take a task which was repetitive, difficult and done using inefficient methods, which could be made substantially better through a software.

- A real-world problem, not a made-up inconvenience
- Something ordinary people would care about

Problem Statement

Multiple university activities independently reserve physical resources and we need some way to detect possible conflicts before people show up, and suggest alternative seating arrangements.

Motivation

Scheduling Conflicts During University Activities

During admission and placement seasons, classrooms are often reserved for online assessments, orientations, and other events.

Regular lectures and lab classes scheduled in these rooms are not always reassigned to alternative available venues.

Faculty members and students may not receive timely notifications about room changes.

This leads to overlapping bookings, last-minute changes, confusion, and disruption of academic activities.

A centralized scheduling system is needed to detect conflicts, suggest suitable alternative rooms, and notify all stakeholders in advance.

Our Solution

01 Conflict Detection

Analyse proposed bookings against existing campus activities, identifying clashes in room, time, capacity, equipment and availability before they occur.

02 Impact Analysis

Determine the consequences of each conflict by identifying affected rooms, activities, teachers and students, allowing the severity and scope of disruption to be understood.

03 Intelligent Resolution

Evaluate available rooms against capacity, equipment, accessibility, distance and existing commitments to recommend the most suitable alternative with minimal disruption.

04 Automated Coordination

Notify affected users of confirmed changes while maintaining an auditable history of bookings, decisions and modifications across the scheduling process.

Core Workflow

Define Requirements : Specify capacity, resources, time slot, activity type, or preferred room.

System evaluates and ranks rooms based on capacity, resources, and schedule.

Coordinator chooses the preferred available venue.

If occupied, system shows the conflicting activity, occupancy, resources, and responsible faculty.

Coordinator decides which activity retains the venue based on priority, timing, and resource needs.

System identifies available venues suitable for the displaced activity.

Coordinator selects an alternative or cancels/reschedules; system records the decision and notifies students and faculty.

The screenshot displays the 'Book a Room' interface under the 'VENUE MANAGEMENT' header. The main title 'Book a Room' is in a large, bold, dark blue font, with a subtitle 'Submit your venue requirements and check availability for your activity.' below it. The interface is divided into two main sections. The left section, titled 'NEW REQUEST' and 'Venue Requirements' (with a '01' badge), contains four input fields: 'ROOM / VENUE' (placeholder: 'Enter room number'), 'ACTIVITY TYPE' (placeholder: 'Lecture, placement examination, orientation...'), 'REQUIRED FACILITIES' (placeholder: 'Projector, computers, microphones...'), and 'REQUIRED CAPACITY' (placeholder: 'Number of students'). A 'DATE' label is at the bottom of this section. The right section, titled 'How it works' (with an 'i' icon), shows a three-step process: 01 'Submit requirements' (Tell us the capacity, facilities and time required.), 02 'Detect conflicts' (Existing venue allocations are checked for conflicts.), and 03 'Find alternatives' (Suitable available venues are recommended.). At the bottom of this section is a green dot and the text 'SYSTEM READY'.

Our frontend

Initial Prototype:

Venue Requirement Form & Conflict Detection:

Capture venue requirements and identify scheduling conflicts.

Post-Conflict Recommendations:

Recommend currently unoccupied rooms that meet the affected activity's capacity, resource, and scheduling requirements.

Public Announcements to display cancellation or relocation notices for affected classes.

Tech Stack

Frontend: CSS, React.js, Vite

Backend: JavaScript, Node.js, Express.js

Database: MongoDB

The screenshot displays the 'CRA CAMPUS RESOURCE ALLOCATOR' interface. At the top right, it says 'RESOURCE MANAGEMENT SYSTEM'. The main content area has a light blue background with a watercolor texture. A green checkmark icon is followed by the text 'VENUE VACANT' and 'LT302 is available to book!'. Below this, there are four white boxes with blue borders, each containing a label and a value: 'REQUESTED VENUE' with 'LT302', 'REQUIRED CAPACITY' with '150', 'DATE' with '2026-08-24', and 'TIME SLOT' with '14:20' and a sub-label '— 15:00'. At the bottom left, there is a blue button with white text that says '← BACK TO REQUEST FORM'.

CRA CAMPUS RESOURCE ALLOCATOR

RESOURCE MANAGEMENT SYSTEM

SUCCESS

VENUE AVAILABLE!

Your requested room is vacant during this time slot.

✓ VENUE VACANT
LT302 is available to book!

REQUESTED VENUE LT302	REQUIRED CAPACITY 150	DATE 2026-08-24	TIME SLOT 14:20 — 15:00
---------------------------------	---------------------------------	---------------------------	--------------------------------------

← BACK TO REQUEST FORM